PHARMACOKINETICS III:
CLINICAL PHARMACOKINETICS

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PHARMACOKINETICS:

Study of absorption, distribution, metabolism and drug excretion over time.

CLINICAL PHARMACOKINETICS:

Application of pharmacokinetic principles to the safe and effective therapeutic management of drugs in an individual patient.

(describes the absorption, distribution, metabolism and elimination of drugs in patients requiring drug therapy)

OBJECTIVES

- Understand the basic concept of clinical pharmacokinetics.
- Understand single dose kinetic and multiple dose kinetics.
- Understand the parameters: C0, Cmax, Css, Tmax, Vd, T1/2, Kel, CL and their clinical significances.

APPLICATION OF CLINICAL PHARMACOKINETICS

- The aim of clinical pharmacokinetics is to ensure the **efficacy** and **safety** of patient's drug therapy.

- Individualization of drug dosage:
- To account for patient variation
- Example: dosage adjustments in neonates, elderly, hepatic and renal impairment

- Pharmacokinetic interactions:
- Example: drugs that interfere drug metabolism, drugs that inhibit drug tubular reabsorption

- Interpret / explain toxicities

Examples:
- Pharmacokinetic drug interactions
- Changes in drug metabolism (e.g. liver impairment)
- Changes in drug excretion (e.g. renal impairment)
- Highly protein binding drugs (e.g. warfarin, phenytoin)

APPLICATION OF CLINICAL PHARMACOKINETICS

- Basis of Clinical Pharmacokinetics
  - there is positive relationship between clinical effects of a drug (therapeutic or toxic), and plasma drug concentration.
  - to produce an optimal therapeutic effect, a "Target Concentration" of drug in the plasma is required.

- Concentration (C) of drug in plasma

- How do you measure??
  - withdraw blood / take blood samples
  - must know **when it is indicated** and **when to take the sample**
  - do not withdraw from the same line as where the drug was administered

- Drug concentration, C is estimated through laboratory analysis (e.g spectrophotometry, chromatography)

- Pharmacokinetic parameters are used to individualize patient's drug therapy.

APPLICATION OF CLINICAL PHARMACOKINETICS

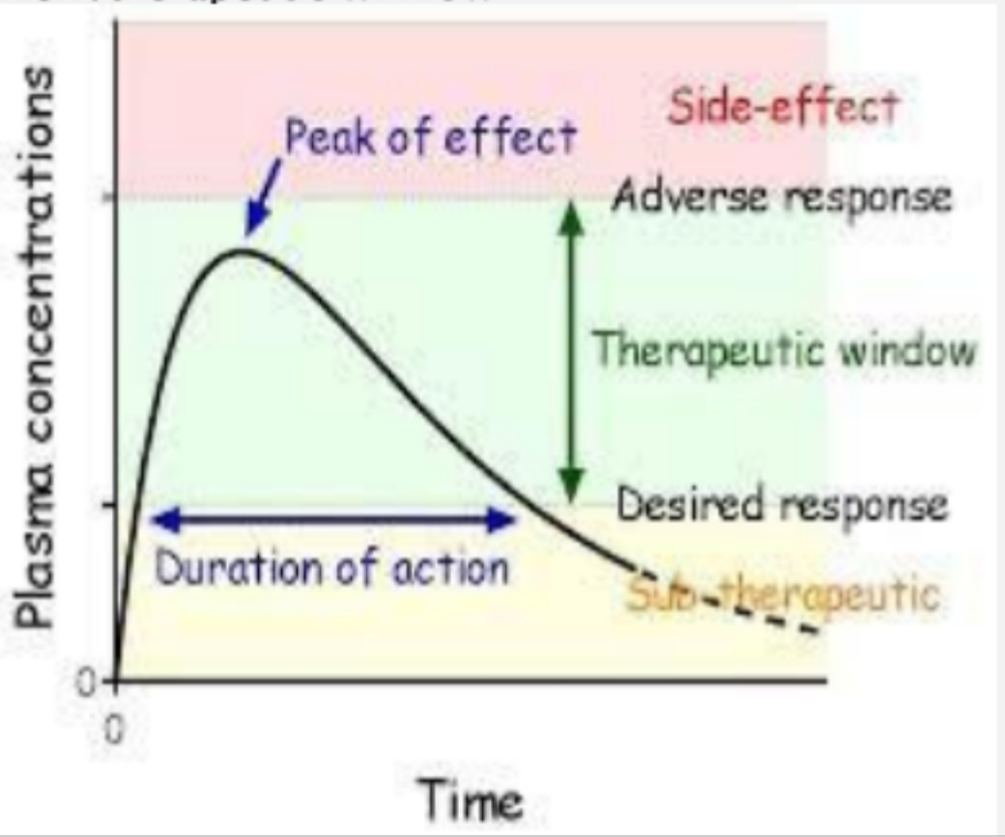
THERAPEUTIC DRUG MONITORING

- Management of patient's drug therapy based on monitoring drug concentrations in the plasma

- When is it indicated?
- The drug has a defined therapeutic range and a narrow therapeutic index /therapeutic window

TABLE 1. Commonly monitored drugs in TDM.

	Class	Drugs
1	Cardiac glycoside ★	Digoxin, Digitoxin
2	Antiarrhythmic	Lidocaine, Quinidine, Procainamide, Disopyramide, Amiodarone, Propranolol
3	Antiasthmatics	Theophylline13
4	Antiepileptic ★	Carbamazepine, Etosuximide, Phenobarbitol, Phenytoin, PrimidoneValoproic acid, Benzodiazepines
5	Antidepressant	Imipramine, Amitriptyline, Clomipramine, Trimepramine, Doxepine, Desipramin, Nortryptyline, Protryptyline, Lithium
6	Antineoplastic	Methotrexate
7	Immunosuppressant	Cyclosporine, Tacrolimus
8	Anticoagulants	Warfarin ★
9	Antibiotics ★	Gentamycin, Amikacin, Tobramycin, Vancomycin14



EXAMPLES OF DRUGS IN TDM

- **Anti-epileptic drug (AED) monitoring**
- To provide a 'benchmark' level in the event the seizure control worsens or toxicity develops
- To assist in investigating breakthrough seizures
- Loading dose required in status epilepticus
- To guide dosage adjustments when switching between generic and brand

- **Vancomycin monitoring**
- For severe infections, critically ill patients, obese patients (BMI > 40kg/m)
- Patients with renal impairment, unstable renal function
- When no clinical response is seen within 3-5 days
- Serum levels to determine frequency of administration (8-24 hourly)

- **Digoxin monitoring**
- Digoxin has a narrow therapeutic index, prone to drug interactions
- Levels need to be check 1 week after starting and regularly afterwards.

DOSAGE REGIMENS

Drugs can be administered as 'one-off' (e.g. painkillers) or at repeated doses (or at different dosage intervals).

1) Single dose kinetics
2) Multiple dosing kinetics
3) Continuous Infusion

DOSAGE REGIMENS

1. Single Dose

- Given 'one off' e.g. painkillers, to induce sleep

- C_0, concentration of drug at time zero

- C_{max}
  - Maximum peak concentration or peak plasma concentration
  - Absorption rate = elimination rate

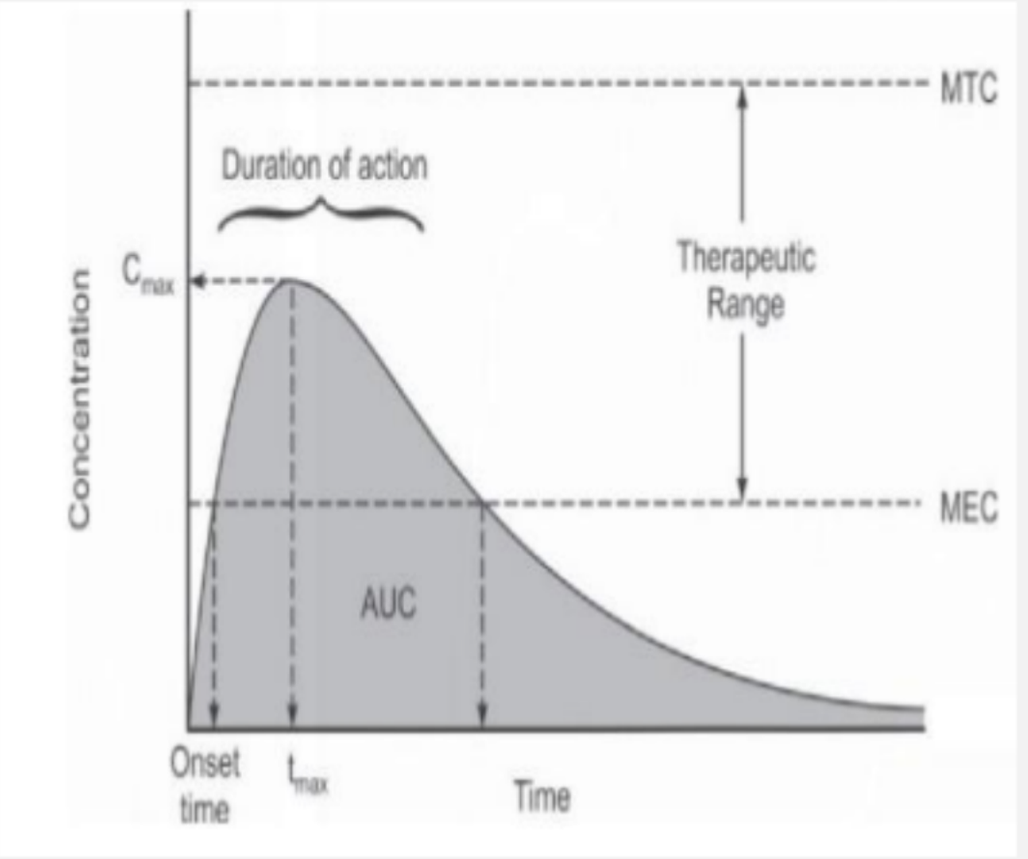
- T_{max}
  - Time to achieve peak concentration, C_{max}

- AUC, Area under the curve
  - Represent total amount of drug exposed to the systemic circulation

- MEC, minimum effective concentration
  - Minimum concentration in plasma to produce a therapeutic effect
  - Drug concentrations below MEC are sub-therapeutic

- MTC, minimum toxic concentration
  - Drug concentrations above MTC is considered toxic

Plasma drug concentration-time Curve



DOSAGE REGIMENS

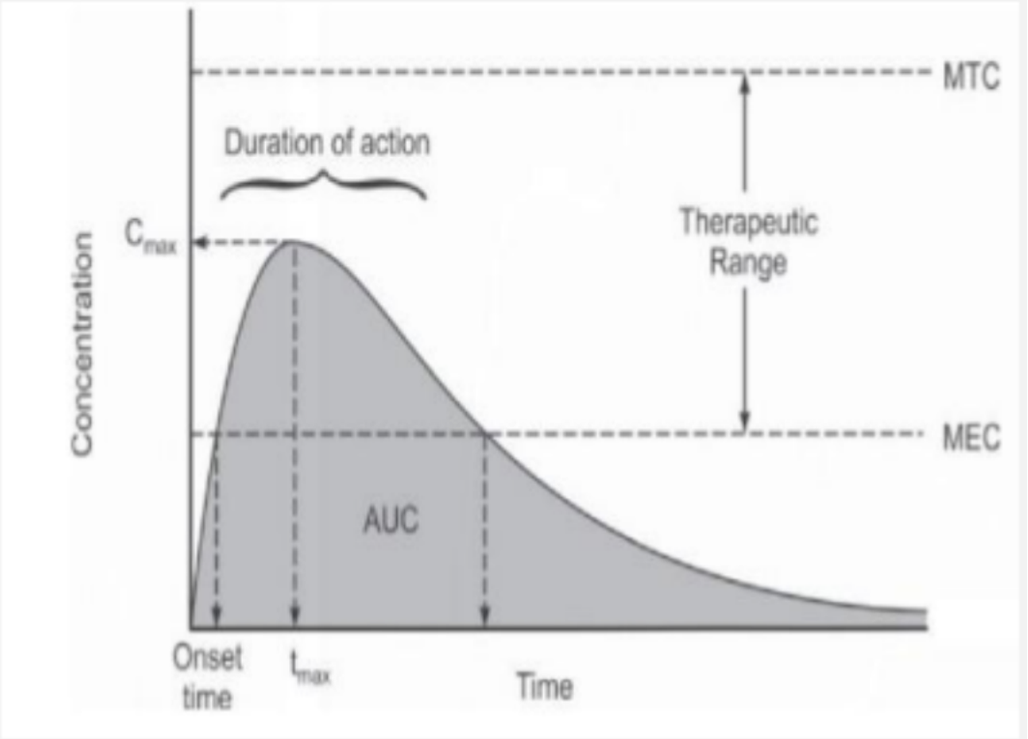
1. Single Dose (cont.)

- Onset time
  - ➢ Time required to exert a therapeutic effect.
  - ➢ Time to reach MEC.

- Duration of Action
  - ➢ Time when drug concentrations are above MEC (duration of drug action)

- Therapeutic Range
  - ➢ Therapeutic window, therapeutic index
  - ➢ Concentration of drug between MEC and MTC
  - ➢ Range of concentration that a drug exerts effectiveness with minimal side effects.

Plasma drug concentration-time Curve

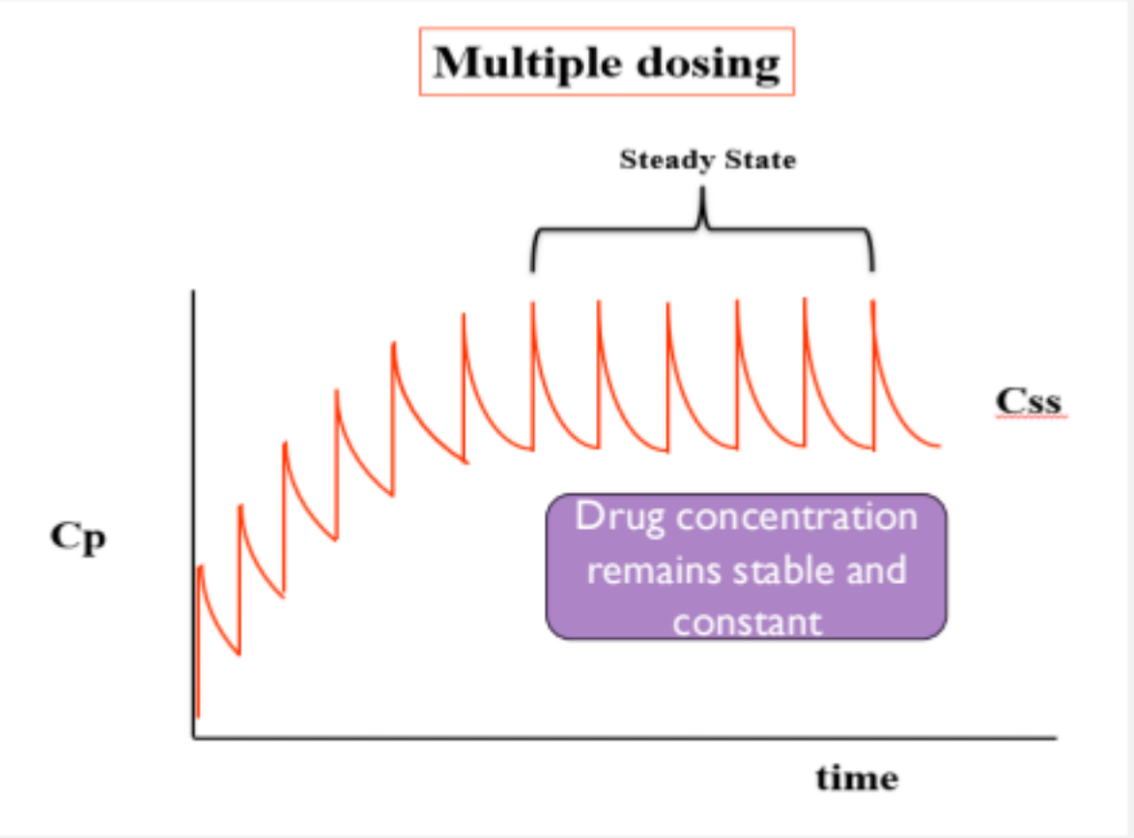


DOSAGE REGIMENS

2. Multiple Dose

- Dose is given repeatedly and results in accumulation:
  - ➢ If the dosage interval is too short, some previous dose will remain in the blood when the second dose is given.
  - ➢ The second peak plasma concentration would be higher than the previous due to accumulation

- Needed when duration of disease is longer than drug in plasma



- **Steady state concentration, C_{ss}**

- when the concentration of the drug in the body stays consistent.
- rate of administration = rate of elimination
- occurs after four to five half-lives of administered drug

- **Maintenance dose, MD**

- the dose needed to maintain steady state concentration

$$MD = \frac{\text{target concentration x clearance x dosing interval}}{F}$$

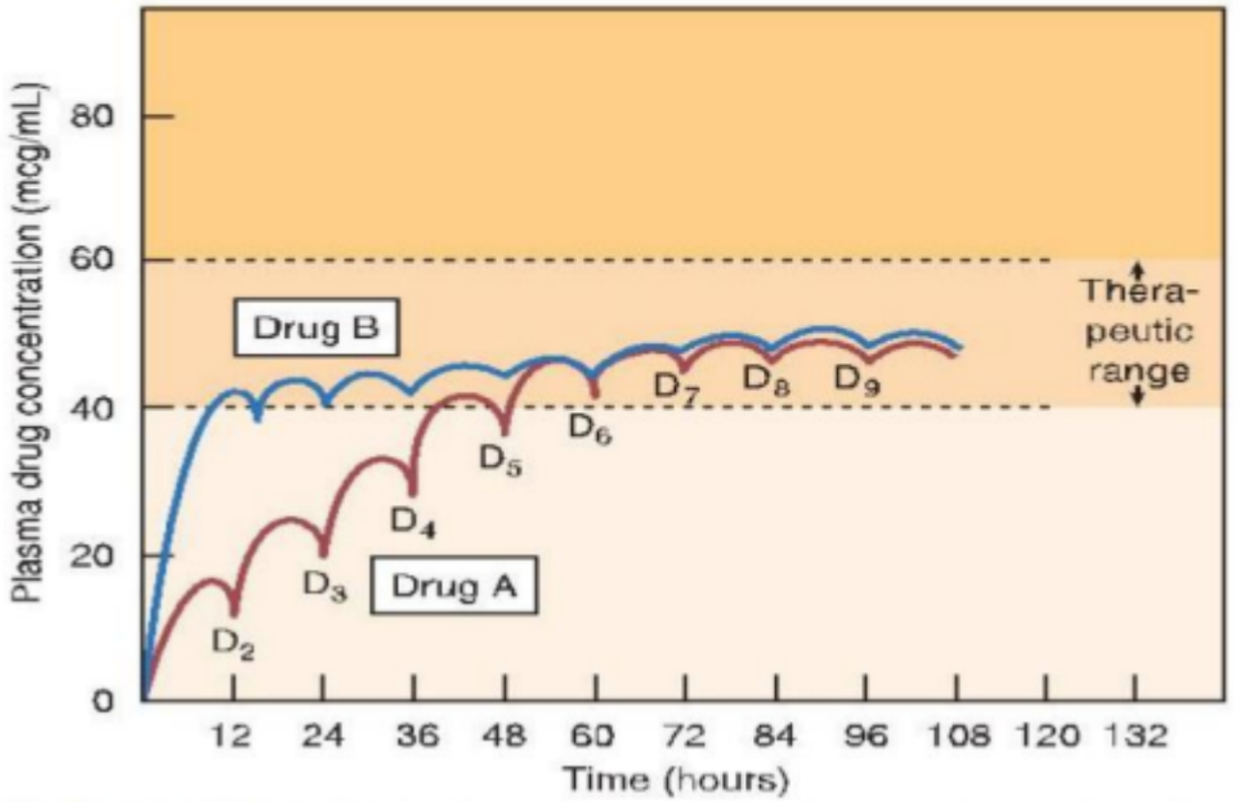
- CL ↓ (renal impairment, hepatic impairment)

- reduce maintenance dose
- reduce dosing interval frequency

- **Loading dose, LD**
- A loading dose may be needed in cases of emergency to achieve the target concentration rapidly
- Example:

Dosage for IV phenytoin in Status epilepticus:
Loading dose of 10-15mg/kg, then a maintenance dose of 100mg q6-8 hourly

Need **Loading Dose** to achieve C_{ss} rapidly



▲ *Figure 4.7* Multiple-dose drug administration: drug A and drug B are administered every 12 hours; drug B reaches the therapeutic range faster, because the first dose is a loading dose

DOSAGE REGIMENS

3. Continuous Infusion

- Drug is entering the circulation at a constant rate
- ***Steady state occurs when the rate of administration equals to the rate of elimination.***
- Drug concentration remains stable and constant.
- C_{ss} – Steady state concentration
- It would take 4-5 half-lives to reach steady state.

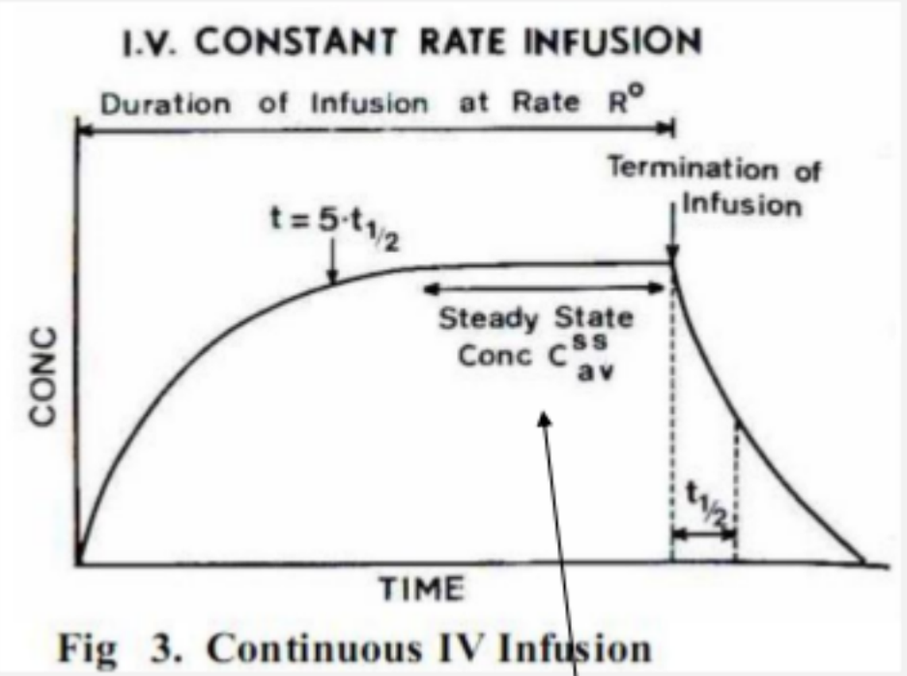


Fig 3. Continuous IV Infusion

Drug concentration remains stable and constant

PHARMACOKINETIC PARAMETERS

ABSORPTION:

- Drugs with faster absorption would have a faster onset because it takes faster to reach the circulation
  - rate of absorption is the amount of drug absorbed per unit time
  - rate is represented by ***absorption rate constant, Kab***, the fraction of administered dose per unit time

- ***Bioavailability (F)*** - *fraction of unchanged drug reaching the systemic circulation*

  - So, if bioavailability is 100%, the fraction (F) is 1. If 80% bioavailability, F = 0.8

  - Bioavailability is used in the calculation of required dose to achieve target concentration.

PHARMACOKINETIC PARAMETERS

DISTRIBUTION:

- ***Volume of Distribution, Vd***

- hypothetical volume into which a drug disperses in order to produce the observed plasma concentration

→ More to representing the proportional relationship between the dose and plasma concentration

- Only for calculation purposes

$$\text{Volume of distribution} = \frac{\text{amount of drug in the body}}{\text{plasma drug concentration}}$$

$$Vd = \frac{\text{Dose} \times F}{C_0}$$

Clinical significance of Vd:

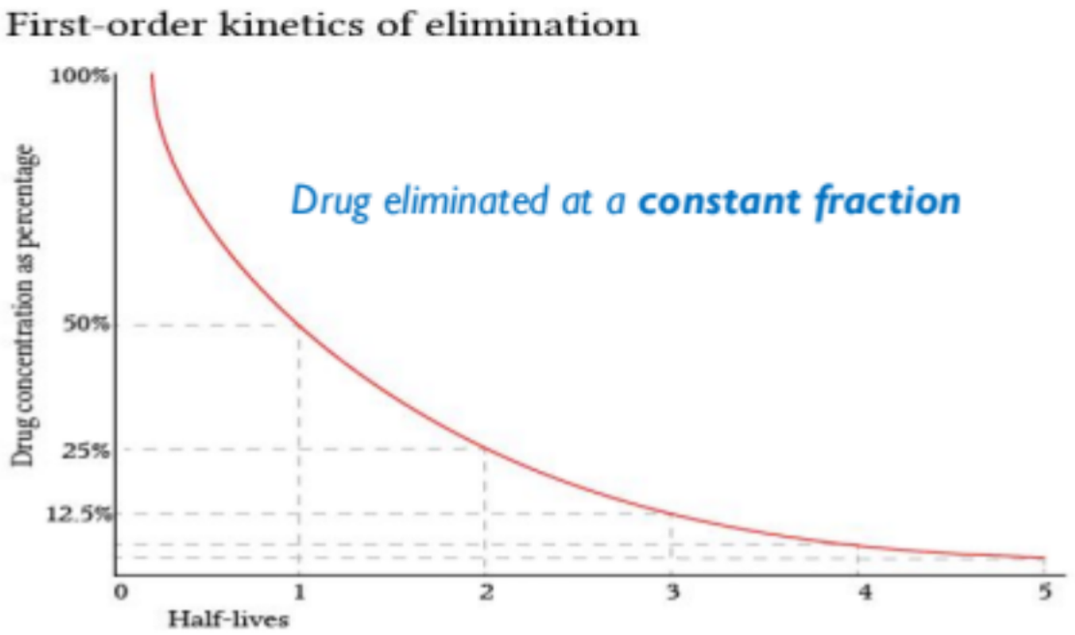
- ↑ Vd means very small amount remains in the blood stream at steady state
- drugs with high Vd is difficult to dialyse during drug poisoning
- heavier person has higher Vd, thus higher dose is needed
- Knowledge of Vd can be used to calculate loading dose:

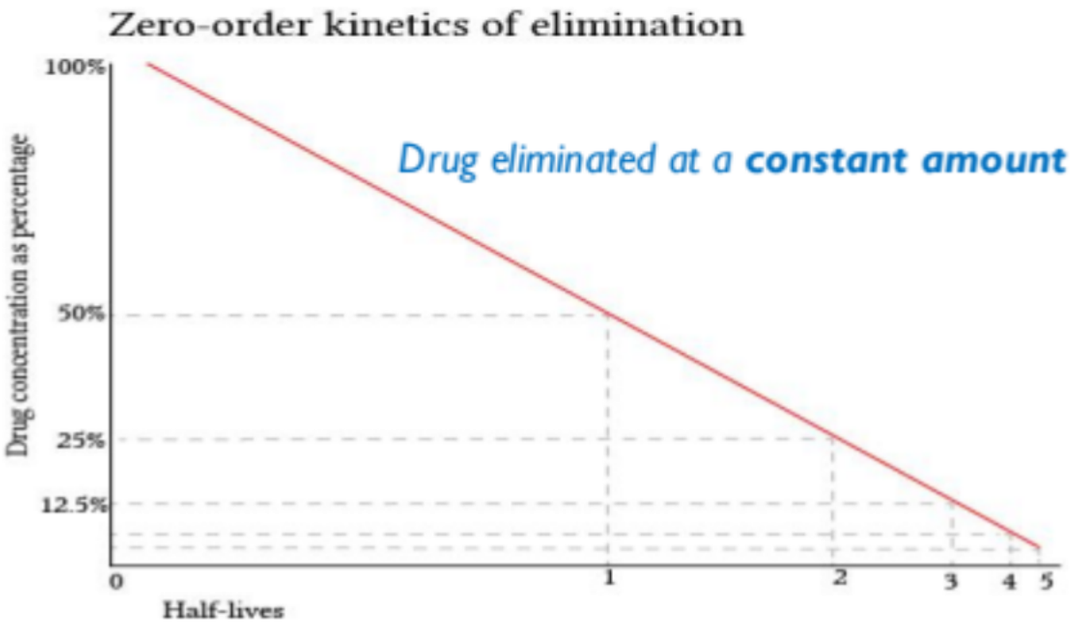
$$LD = \frac{\text{target concentration} \times Vd}{F}$$

PHARMACOKINETIC PARAMETERS

ELIMINATION

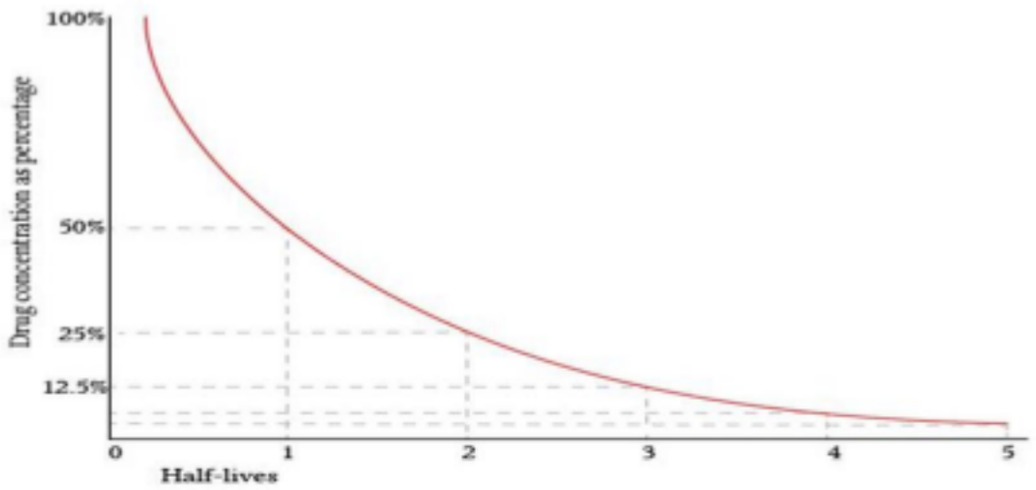
- Elimination is the MOST important factor that influences plasma drug concentration
- Drugs can follow **First-Order Elimination** (most drugs) or **Zero-Order Elimination** (phenytoin, aspirin)



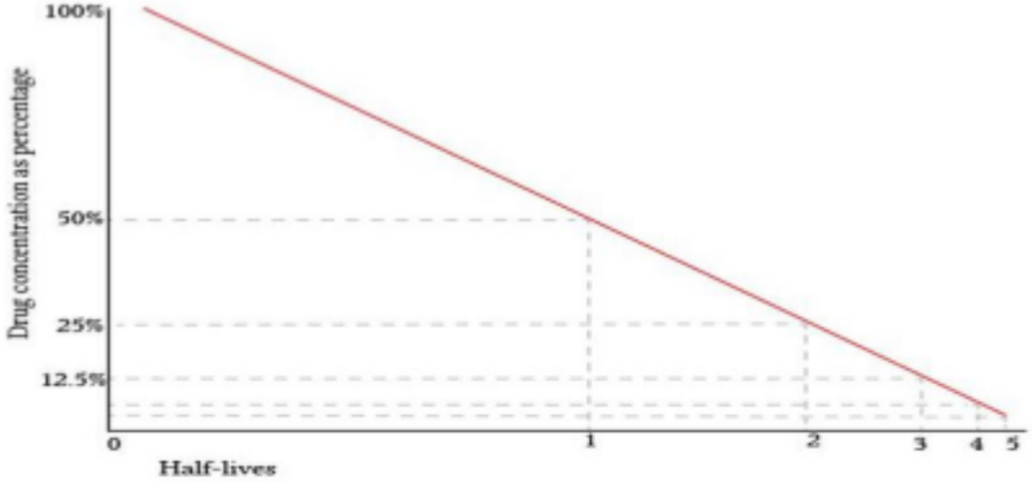


- Half-Life ($t_{1/2}$) – time needed for drug concentration to drop by 50%

First-order kinetics of elimination



Zero-order kinetics of elimination



Comparison

- **First Order Elimination**
  - [drug] decreases **exponentially** w/ time
  - Rate of elimination is proportional to [drug]
  - Plot of log [drug] or ln[drug] vs. time are linear
  - $t_{1/2}$ is constant regardless of [drug]
- **Zero Order Elimination**
  - [drug] decreases **linearly** with time
  - Rate of elimination is constant
  - Rate of elimination is independent of [drug]
  - No true $t_{1/2}$

Prone to toxicities due to accumulation or when elimination process is saturated.

PHARMACOKINETIC PARAMETERS

- *Clinical significance of half-life, $t_{1/2}$:*

- Can predict when the drug is fully eliminated as most of the drug is eliminated at about 4-5 half-lives (e.g. when to initiate drug during switching, acute poisoning cases)

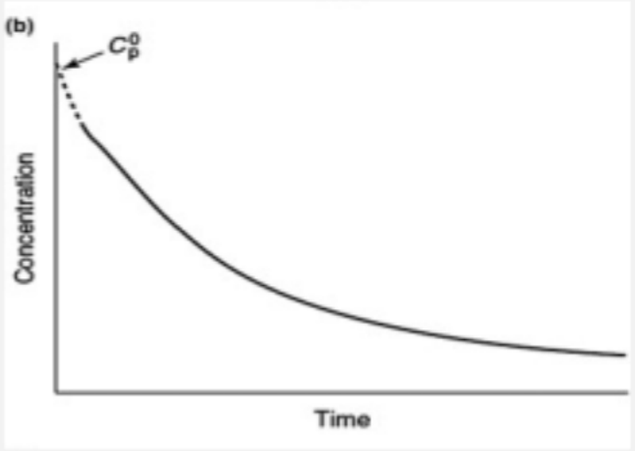
- determine dose regimens. (E.g. Drugs with short half-lives would need more frequent dosing)

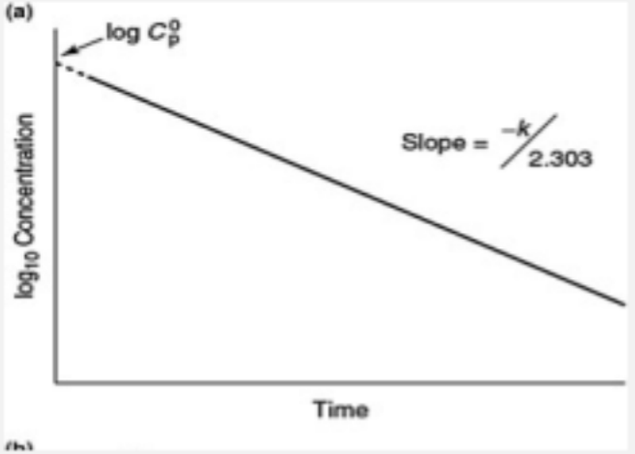
- used in therapeutic drug monitoring (TDM)

PHARMACOKINETIC PARAMETERS

- Elimination rate constant, K_{el}:
- the fraction of drug in the plasma being eliminated per unit time

- Calculated from concentration data (slope)





$$K_{el} = \frac{\text{natural log } 2}{T_{1/2}} = \frac{0.693}{T\,½}$$

- for drugs that follow first order elimination kinetics
- inversely proportional to half-life
- used to calculate clearance, CL

$$CL = Vd \times Kel$$

PHARMACOKINETIC PARAMETERS

ELIMINATION

- ***Clearance***

➢ the rate at which a drug is removed from plasma(mg/min) divided by the concentration of that drug in the plasma (mg/mL).

- **Total clearance = renal clearance + hepatic clearance + clearance from all other tissues.**

- Represents elimination from the body as a whole



Clinical significance of clearance:

- Determines the maintenance dose required to reach steady state concentration

- Maintenance Dose:

$$\frac{\text{target concentration} \times \text{clearance} \times \text{dosing interval}}{F}$$

- CL ↓ (renal impairment, hepatic impairment)

- reduce maintenance dose
- reduce dosing interval frequency

CLINICAL SIGNIFICANCES OF CLINICAL PHARMACOKINETICS

Clinical Significances of clinical pharmacokinetics

1. Calculations employing principles of clinical Pharmacokinetics can find out:
(a) Plasma drug concentration which can be achieved by a particular dosage regimen
(b) whether the plasma drug level achieved is therapeutic, subtherapeutic or toxic
(c) Dosage regimen required in an individual to achieve target concentration

2. Contributes to making decision in drug therapy.
- E.g. whether to adjust dosage or switch to another drug, to achieve optimal therapeutic outcome

SUMMARY OF PHARMACOKINETIC PARAMETERS & FORMULA

Absorption : F, Kab

Distribution : Vd

Elimination : $t_{1/2}$, k_{el}, CL

$$Vd = \frac{Dose \times F}{C_0}$$

$$K_{el} = \frac{0.693}{T\,1/2}$$

$$CL = Vd \times Kel$$


THANK YOU!